

Real Estate Property Data Cleaning and Analysis

A Data Cleaning and Analysis Report Using Microsoft Excel

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1. Introduction

This report presents a data cleaning and analysis project carried out on a real estate property dataset using Microsoft Excel. The dataset contains information about properties listed for sale, including their type, location, size, number of bedrooms, listing date, and price.

Before the data could be analyzed, it needed to be cleaned. The original data had a number of problems, such as missing values, duplicate records, inconsistent text formatting, and a few invalid entries. These issues had to be fixed first, because analysis based on messy data would give misleading results.

Once the data was cleaned, it was analyzed to understand the property market: which locations have the most listings, which property types are most common, how prices differ across locations, and how prices are distributed overall. The purpose of this project is to demonstrate a complete data cleaning and analysis workflow in Excel, from identifying problems in raw data to producing clear, reliable findings.

2. Dataset Overview

The raw dataset is stored in the Raw_Data sheet. It contains 300 records and 7 main columns describing each property listing:

Column	Description
Property_ID	A unique code identifying each property listing (for example, PRP0045).
Listing_Date	The date on which the property was listed for sale.
Property_Type	The category of property, such as Apartment, House, Villa, Land, or Commercial.
Location	The city where the property is located (Mogadishu, Hargeisa, Bosaso, Baidoa, Garowe, or Kismayo).
Size_Sqm	The size of the property, measured in square meters.
Bedrooms	The number of bedrooms in the property.
Price	The listed price of the property.

3. Problems Found in the Raw Data

A close inspection of the Raw_Data sheet showed several data quality problems. These are summarized column by column below.

Column	Problem Found	Amount / Example	How It Was Fixed
Property_ID	Missing IDs; duplicate IDs	15 missing; 8 duplicate IDs (e.g. PRP0032, PRP0046)	Missing IDs replaced with placeholders (Unknown01-Unknown15); duplicate rows removed, keeping one record per ID
Listing_Date	Missing dates; one invalid entry	15 missing; 1 entry stored as "-"	Missing or invalid dates replaced with the text "Unknown"; valid dates kept as they were
Property_Type	Missing values; "-" placeholder; inconsistent capitalization	17 missing/blank; 1 "-"; 5 rows with mixed case (e.g. house, HOUSE, land)	Missing/blank/"-" entries set to "Unknown"; text standardized to proper case (Apartment, House, Villa, Land, Commercial)
Location	Missing values; inconsistent capitalization	18 missing/blank; 6 rows with mixed case (e.g. GAROWE, garowe, MOGADISHU)	Missing/blank entries set to "Unknown"; text standardized to proper case (Mogadishu, Garowe, Hargeisa, Baidoa, Kismayo)
Size_Sqm	Missing values	15 missing	Missing values filled with the average size of the valid records (about 295 sqm)
Bedrooms	Missing values; "N/A" text entries; negative values	16 missing; 3 "N/A" entries; 2 negative values (-3, -7)	Missing/"N/A" entries filled with the median value (3 bedrooms); negative values corrected to their positive equivalents
Price	Missing values; one "NULL" text entry; one negative value	15 missing; 1 "NULL" entry; 1 negative value (-6)	Missing/"NULL" entries filled with the average price of the valid records (about \$178,260); the negative value was also corrected using this average

In total, the Raw_Data sheet contained 111 missing values across all seven columns, based on the data quality checks in the Analysis sheet.

4. Data Cleaning Methodology

The data was cleaned following a clear, step-by-step process:

- Raw Data - The original 300-record dataset was reviewed first.
- Identify Problems - Each column was checked for missing values, duplicates, wrong data types, and inconsistent text.
- Clean Missing Values - Missing numeric values (Size_Sqm, Bedrooms, Price) were filled using averages or medians; missing text values (Property_ID, Listing_Date, Property_Type, Location) were replaced with clear "Unknown" placeholders.
- Remove Duplicates - The 8 duplicate Property_ID records were removed, keeping only one record per property.

- Standardize Text - Property_Type and Location text was converted to proper case using the PROPER function so that entries like "house" and "HOUSE" were treated the same way.
- Validate Data - Negative or otherwise invalid values in Bedrooms and Price were corrected.
- Clean Data - The final, cleaned dataset was produced in the Clean_Data sheet, ready for analysis.

The results of this process, according to the Analysis sheet, were as follows:

Quality Check	Before Cleaning	After Cleaning
Total Records	300	292
Duplicate Records	8	0
Missing Values	111	0

5. Before Cleaning vs After Cleaning

Before Cleaning

- 300 records in total
- 8 duplicate records
- 111 missing values across all columns
- Inconsistent text values (mixed capitalization in Property_Type and Location)
- Missing Property_IDs, listing dates, property types, locations, sizes, bedroom counts, and prices

After Cleaning

- 292 records in total
- 0 duplicate records
- 0 missing values - all gaps were filled or marked as "Unknown"
- Text values standardized to consistent proper case
- 7 additional calculated columns added for analysis (price category, days since listing, extracted ID, location length, standardized type, standardized location, and year)

The cleaned dataset is more reliable for analysis because every record has complete information, duplicate listings no longer inflate the counts, and text values are consistent, so properties can be grouped and compared accurately by type and location.

6. Formulas and Excel Functions Used

The Clean_Data sheet uses several Excel formulas to prepare the data for analysis. These are explained below.

Price Category

Formula: =IFS(Price>=200000,"High",Price>=100000,"Medium",Price<100000,"Low")

This formula sorts each property into a High, Medium, or Low price category based on its price. Properties priced at \$200,000 or more are High, those from \$100,000 up to just under \$200,000 are Medium, and anything under \$100,000 is Low. This makes it easy to see how many properties fall into each price bracket.

Days Since Listing

Formula: =IF(OR(Listing_Date="Unknown",Listing_Date=""),"Unknown",TODAY()-Listing_Date)

This formula calculates how many days have passed since a property was listed, using today's date. If the listing date is missing or marked "Unknown", the formula returns "Unknown" instead of an error.

ID_Extracted

Formula: =RIGHT(Property_ID,4)

This formula uses the RIGHT function to pull out the last 4 characters of each Property_ID (the numeric part), which is useful for sorting or referencing properties by their ID number.

Location Length

Formula: =LEN(Location)

This formula uses the LEN function to count the number of characters in the Location text. It is a simple data-quality check that can help spot unusually short or long entries.

Standardized Property Type

Formula: =PROPER(Property_Type)

This formula uses the PROPER function to convert Property_Type text to proper case (for example, "house" and "HOUSE" both become "House"), so that identical categories are not treated as different values.

Standardized Location

Formula: =PROPER(Location)

This formula works the same way as above but is applied to the Location column, standardizing entries such as "GAROWE" and "garowe" into "Garowe".

Year Column

Formula: =IF(Listing_Date="Unknown","Unknown",YEAR(Listing_Date))

This formula uses the YEAR function to extract the listing year from the Listing_Date column, which is what makes it possible to compare the number of properties listed in 2025 versus 2026. Records with an unknown date are labeled "Unknown" rather than causing an error.

7. Data Analysis

After cleaning, the dataset was analyzed using the formulas in the Analysis sheet and the pivot tables in the PivotTables sheet. The main results are summarized below.

Metric	Value
Total Properties	292
Total Property Value	\$52,230,254.08
Average Price	\$178,870.73
Minimum Price	\$17,505.31
Maximum Price	\$2,000,022.00
Average Property Size	294.53 sqm
Average Bedrooms	2.63
Properties Listed in 2025	169
Properties Listed in 2026	107
Most Common Property Type	Apartment
Top Location	Mogadishu

The price categories created earlier also show that 85 properties fall into the High price bracket, 101 into Medium, and 106 into Low, giving a fairly balanced spread across the three price levels.

8. Analysis by Location and Property Type

The cleaned dataset covers six named locations plus a small group of records where the location was originally unknown. The table below shows the number of properties and the average price for each location.

Location	Number of Properties	Average Price
Mogadishu	101	\$209,588.11
Hargeisa	51	\$157,337.25
Bosaso	36	\$153,062.86
Baidoa	35	\$146,560.47
Garowe	30	\$206,753.68
Kismayo	20	\$133,616.89
Unknown	19	\$185,411.96

Mogadishu clearly stands out, with the highest number of properties in the dataset and one of the highest average prices, just ahead of Garowe. Hargeisa, Bosaso, and Baidoa form a middle group with moderate listing counts, while Kismayo has the fewest listings and the lowest average price.

Looking at property types, Apartment is the most common category with 96 properties, followed by House (59), Land (46), Villa (37), and Commercial (34), with a further 20 records marked "Unknown". In terms of total value, however, Land properties account for the largest combined price (\$13,521,302.71), followed by Villa (\$9,803,140.87) and Apartment (\$9,587,232.32), reflecting the generally higher price of land and larger properties.

9. Findings

- Mogadishu has the highest number of properties, with 101 listings, making it the largest market in the dataset.
- Apartment is the most common property type, with 96 properties, followed by House with 59.
- Mogadishu has the highest average property price, at about \$209,588, slightly ahead of Garowe at \$206,754.
- Data quality improved significantly after cleaning: 8 duplicate records were removed and all 111 missing values were handled, leaving 292 reliable, complete records.

10. Recommendations

- Focus more on the Mogadishu property market, since it has both the largest number of listings and strong average prices.
- Monitor demand for apartments, as they make up the largest share of available properties.
- Review property prices by location, especially in Mogadishu and Garowe, to better understand why prices are higher there.
- Keep the dataset clean and updated regularly, to reduce missing, duplicate, and unknown information going forward.

11. Conclusion

Overall, this project shows that cleaning the raw real estate data made the dataset far more reliable for analysis. The original 300 records contained 8 duplicates and 111 missing values, but after the cleaning process the dataset was reduced to 292 complete, duplicate-free records with standardized text throughout. The analysis of this cleaned data shows that Mogadishu has both the largest number of properties and one of the highest average prices, while apartments are the most common property type across the market. Together, these results demonstrate the value of proper data cleaning: the cleaned dataset provides a much stronger and more trustworthy basis for understanding the property market and making informed decisions.